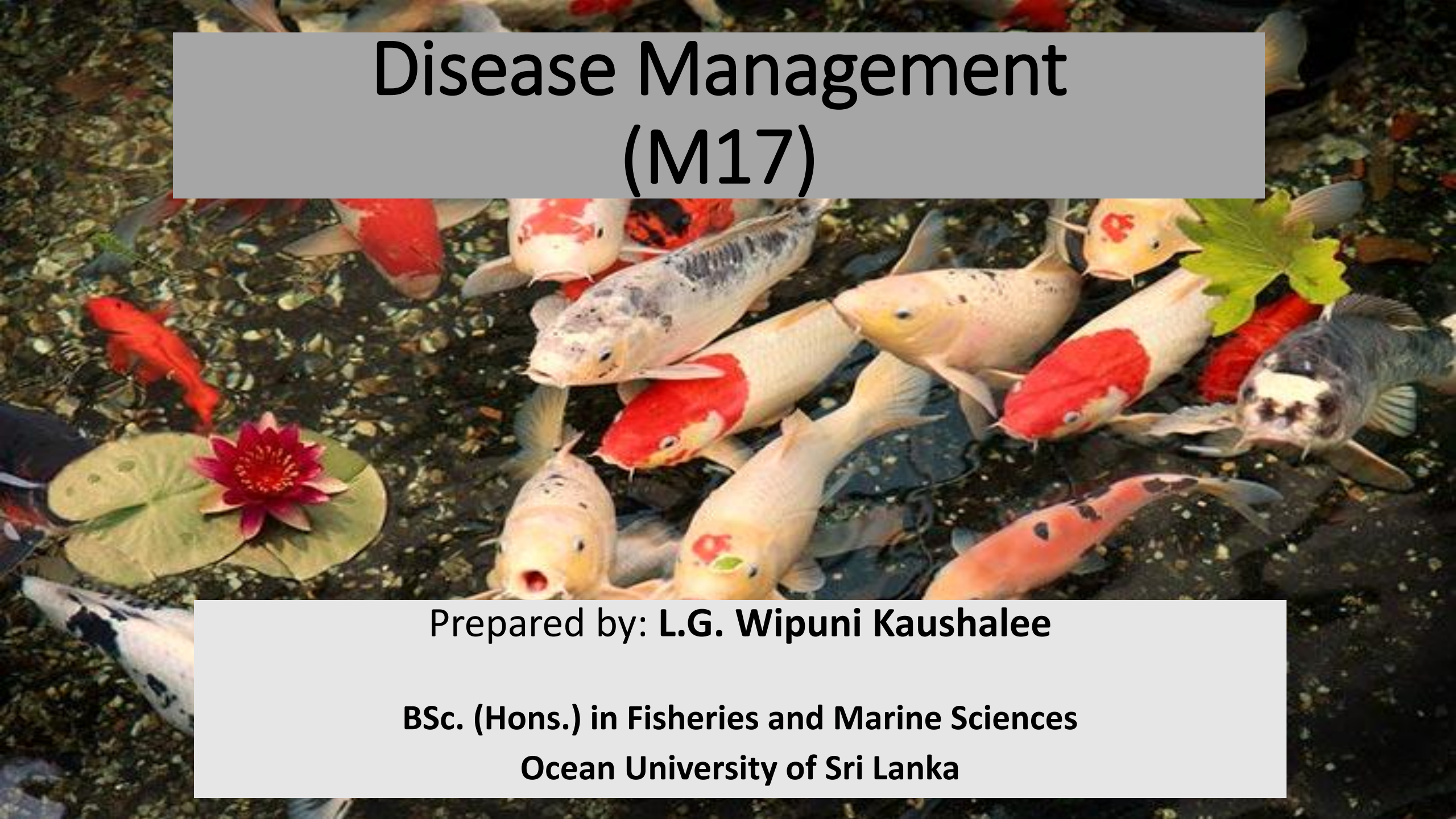




Disease Management (M17)

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Learning Outcomes

- Demonstrate the knowledge on diagnosis of common diseases Identify, treat and control diseases.
- Apply prevention methods.



2. Bacterial Diseases

- Diseases caused by bacteria may cause heavy mortality in both wild and cultured fish and crustaceans.
- Bacteria are found everywhere in the aquatic environment.
- Most bacterial disease agents are part of the normal microflora of the marine environment and are generally considered as secondary or opportunistic pathogens.
- Almost all fish bacterial pathogens are capable of independent existence outside the fish.

- There are only a few obligatory pathogens.
- Even these, however, are capable of living for a long time in the tissues of their host without causing injury.
- Clinical infections and disease usually occur only after the onset of some major changes in the physiology or body of the host.
- Thus, to understand bacterial diseases of fish, one must understand the relationship of bacteria with their host and with their environment.

- As in all animal production systems, bacterial disease is one of the major problems facing production, development and expansion of the aquaculture industry.
- The control of disease is particularly difficult because fish are often farmed in systems where production is dependent on natural environmental conditions.
- Changes or deterioration in the aquatic environment cause most of the bacterial diseases encountered, and environmental effects give rise to many other adverse conditions.

- A second major constraint on disease control is the relatively limited number of therapeutic agents available for the control of bacterial disease agents.
- Even recommended, therapies and preventive measures pose limitations.
- Their application to aquatic animals is often difficult in actual practice, and sometimes impossible to implement.

- Most bacterial disease show similar signs, especially in fishes. Bacterial infection may appear on the skin or fins of fish, exoskeleton or appendages of crustaceans, in the muscles and in the internal organs.
- In nearly all cases, red spots, brown or black spots, or necrotic tissues can be observed.
- Inflammation may also occur.
- Proper identification of the causative agent is important to ensure successful treatment.

Prevention of Bacterial Diseases

- Outbreaks of major bacterial diseases in aquaculture can be significantly reduced if proper attention is paid to good husbandry practices and the maintenance of optimum environmental conditions, especially water quality.
- Another important consideration involves identifying the predisposing factors that may lead to a disease state.
- Once predisposing factors are identified, appropriate corrective measures should be initiated in the culture system.

1. Columnaris Disease

CAUSATIVE AGENT: *Flavobacterium columnare* (previously named as *Flexibacter columnaris*)

SPECIES AFFECTED: Tilapia (*Oreochromis niloticus*), carp (*Cyprinus* sp.), channel catfish (*Ictalurus punctatus*), goldfish (*Carassius auratus*), rohu (*Labeo rohita*)



GROSS SIGNS:

- The first indication of infection is generally the appearance of a white spot on some part of the head, gills, fin or body.
- A zone with a distinct reddish tinge usually surrounds this.
- The lesions are circular as if spreading from a single focus towards all directions at the same rate.
- On the gills, the lesions are more necrotic.
- On the skin, they develop into hemorrhagic ulcers, with an overlying seroma of bacterial cell and necrotic tissue.
- Histologically, there is epidermal spongiosis, necrosis, and ulcerations.

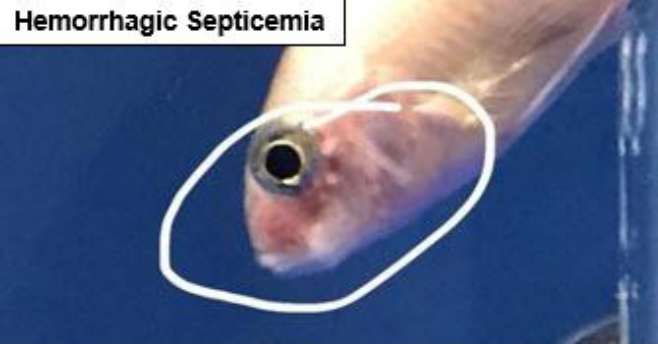
EFFECTS ON HOST:

- *F. columnare* is an opportunistic pathogen widely distributed in the water.
- The disease does not usually occur as spontaneous infection but results from injuries to the fish, or physical and nutritional deficiencies.
- Outbreaks are affected by factors such as temperature and stress.
- Dissolved cations such as sodium, potassium, calcium and magnesium enhance their infectivity.
- *F. columnare* attacks fish primarily through the gills or abraded epidermal areas.

- Once the pathogen is established, proteolytic enzymes break down the skin and muscle to open necrotic lesions.
- The bacterium appear systematically after extensive tissue necrosis.
- Gill lesions may cause respiratory difficulty and the fish eventually dies.
- Body lesions are subject to secondary infection by other microorganisms.
- Fish that survive the infection may become carriers of the disease.

DIAGNOSIS:

- Columnaris disease can be presumptively diagnosed from disease signs on the skin and gills of the host and from squash preparations made from scrapings of the affected areas.
- In wet mount preparations of infected tissues, the bacteria show a slow gliding movement, gathering into characteristic column-like masses that give the disease its name.
- Microscopic examination of lesions shows the presence of long, slender, possibly filamentous, rod-shaped, gram-negative bacteria.
- The growth of *F. columnare* on solid media is usually characterized by yellow-green, flat and rough spreading colonies that adhere to the media.
- *F. columnare* can also be detected from fish and water using indirect fluorescent antibody technique (IFAT).



2. Pseudomonad Septicemia or Red Spot Disease

CAUSATIVE AGENT: *Pseudomonas fluorescens*, *P. anguilliseptica*, and *P. chlororaphis*.

SPECIES AFFECTED: Milkfish (*Chanos chanos*), goldfish (*Carassius auratus*), tilapia (*Oreochromis niloticus*)

GROSS SIGNS:

- The external disease signs of pseudomonad septicaemia are similar to those caused by other gram-negative bacterial pathogen of fish.
- The disease causes small hemorrhages in the skin around the mouth and opercula and along the ventral or abdominal surfaces.
- The body surface may ooze blood and slime in severe cases but there is no reddening of the fins and anus.

EFFECTS ON HOST:

- *Pseudomonas* spp. enters the host either through the oral route or through broken or abraded skin and damaged gills.
- The organism is carried throughout the fish body by the blood stream.
- The bacteria and their toxin destroy body tissues, organs and functions.
- Dysfunctions of the different body organs may lead to mortality of up to 70%.

DIAGNOSIS:

- The organisms can usually be isolated from the kidney and other internal organs of affected fish, as well as from the lesion.
- They grow well on most common laboratory media such as TSA (Tryptic Soy Agar) and NA (Nutrient Agar).
- GSP (Glutamate Starch Phenol Red) agar or *Pseudomonas-Aeromonas* Selective Agar is a medium that selectively promotes growth of *Aeromonas* and *Pseudomonas*, and inhibits growth of other bacteria.

PREVENTION AND CONTROL:

- Maintain proper stock management procedures, ensure water quality and reduce stocking density.
- Transfer in a tank and raise temperature to 26-27°C and maintain for about 2 weeks.

3. Mycobacteriosis or Piscine Tuberculosis

CAUSATIVE AGENT: *Mycobacterium marinum*, *M. fortuitum* and *M. chelonae*

SPECIES AFFECTED: Siamese fighting fish (*Betta splendens*)



GROSS SIGNS:

- Piscine mycobacteriosis is a systemic, chronic, progressive disease presenting various clinical features depending upon species and ecological conditions.
- Listlessness, anorexia, emaciation, exophthalmia, skin discoloration and external lesions ranging from scale loss to nodules, ulcers and fin necrosis are signs of advancing infection.
- Gross internal pathology of mycobacteriosis show graywhite lesions of various sizes in most organs and tissues, but the kidney and liver are most often involved.

EFFECTS ON HOST:

- Mycobacteriosis is a chronic progressive disease.
- The disease may take several years to progress from the asymptomatic state to clinical illness.
- Initially the pigment will fade, and the fish appear sluggish with loss of appetite.
- Skin ulcers will develop.
- Fin and tail rot and loss of scale may also be seen.
- Nodules may form in the muscle and internal organs, which may lead to emaciation or edema or peritonitis.
- Infection may spread to the skeleton, in which deformities become apparent.
- Mortalities will then be observed.

DIAGNOSIS:

- Primary isolation of fish mycobacteria is best achieved using Ogawa and Lowenstein-Jensen media.
- Subcultures develop at 28°C within 3-5 days on these media.
- On Ogawa medium, the cultures appear creamy in the dark but brilliant yellow color when exposed to light.
- Cultures may not always be obtained even from fish showing unequivocal evidence of infection.

- Mycobacteria may also be isolated occasionally on general-purpose bacteriological media such as tryptic soy agar, or brain heart infusion agar, provided that a large inoculum is used.
- All fish mycobacterium have been cultured at 20-30°C for 2 to 30 days.
- The isolates are strongly acid-fast, rod-shaped, weakly gram-positive, cord forming, non-motile and non-spore forming.
- Optimum temperature for bacterial growth is between 15°C to 37°C, but the isolates grow best at 28°C.

PREVENTION AND CONTROL:

- Sanitation, disinfection, and destruction of carrier fishes are the primary methods of controlling mycobacteriosis.
- Avoid feeding fish with contaminated fish products. Pasteurize food before use.
- Apply chloramine B or T at 10 mg/l for 24 h.

Bacterial Diseases Of Crustaceans

- Bacterial infections of cultured crustaceans occur as:
 - ✓ Bacterial fouling of surfaces
 - ✓ Cuticular or subcuticular localized infections
 - ✓ Internal or systemic infections

4. Bacterial Fouling of Surfaces: Filamentous Bacterial Disease

CAUSATIVE AGENT: *Leucothrix* sp., *Thiothrix* sp., *Flexibacter* sp., *Cytophaga* sp., *Flavobacterium* sp.

SPECIES AFFECTED: *Penaeus monodon*, *P. merguensis*, *P. indicus*

GROSS SIGNS:

- Presence of fine, colorless, thread-like growth on the body surface and gills as seen under a microscope.



Filamentous Bacteria on Shrimp Body



Necrosis and inflammation of the Shrimp tail

EFFECTS ON HOST:

- Infected eggs show a thick mat of filaments on the surface, which may interfere with respiration or hatching.
- In larvae and postlarvae, normal respiration, feeding, locomotion, and molting may be seriously impaired, resulting in slower growth rates, retarded development and eventually death.
- However, larval shrimps are less prone to infestations due to rapid succession of molts throughout the different larval stages.
- Frequent molting does not allow adequate time for the bacteria to accumulate on the exoskeleton.

- In larger shrimps, it may result in respiratory distress.
- Mortality is due to hypoxia.
- Disease onset is associated with high organic loads in culture water, low dissolved oxygen levels and added stress from molting.
- If left untreated in intensive culture systems, accumulative mortality may reach 80% or more within a few days to a few weeks of onset of disease signs.

DIAGNOSIS:

- Direct microscopic examination of wet mounts of larvae or postlarvae, appendages and gill filaments excised from juvenile or adult shrimp, and of filamentous organisms attached to external surfaces of the cuticle.

PREVENTION AND CONTROL:

- Maintain good water quality with optimum dissolved oxygen and low organic matter levels.
- Apply Cutrine Plus at 0.15 ppm copper in 24 h flowthrough treatments.
- Apply 0.5 ppm copper in 4 to 6-h static treatments for PL 2 and older.

5. Cuticular or Subcuticular Localized Infections: Shell Disease, Brown/Black Spot, Black Rot/Erosion, Blisters, Necrosis of Appendages

CAUSATIVE AGENT: Shell-degrading bacteria
belonging to *Vibrio*, *Aeromonas*, and *Pseudomonas*
groups

Species affected: *Penaeus monodon*, *P. indicus*



Figure 3-12a. Shell disease on the carapace of *Penaeus monodon*

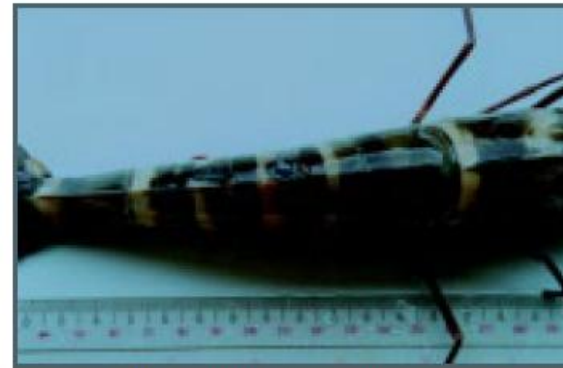


Figure 3-12b. Shell disease on the abdominal segment of *Penaeus monodon*



Figure 3-13. *Penaeus monodon* post larvae with necrotic pleopods. Necrotic area appears like cigarette butt

GROSS SIGNS:

- The disease manifests itself as brownish to black, single or multiple, eroded areas on the general body cuticle, appendages, and gills.
- In larval and postlarval stages, the affected appendage shows a cigarette butt-like appearance.
- Blister containing cyanotic gelatinous fluid may develop on the carapace and abdominal segment.
- The blister may extend to the underside of the ventro-lateral section of the carapace creating a bulge on the underside.

EFFECTS ON HOST:

- Infection usually starts at sites of punctures or injuries caused by the telson or rostrum, in cracks on the abdominal segment from sudden flexure of the shrimp body, or from other damage caused by cannibalism.
- Another infection site is the cuticle colonized by a large number of bacteria.
- The bacteria produce extracellular lipases, proteases, and chitinases, which together erode the multiple-layered cuticle, resulting in the development of the disease.
- The progressive destruction of the cuticle also provides a route of entry for secondary pathogens like fungi or opportunistic bacteria.

- Such infections may become lethal because of osmotic imbalances, molting problems, secondary fungal infection and a generalized septicemia.
- The affected shrimp becomes susceptible to cannibalism or dies from stress and energy exhaustion.
- The disease is associated with trauma to the cuticle (e.g. heavy aeration), conditions that encourage a high number of bacteria in the culture water (e.g. poor hatchery hygiene, high organic loads or contaminated algae) and undefined nutritional and environmental stressors.

DIAGNOSIS:

- Diseased penaeids are examined for appearance of multifocal melanized cuticular lesions on the cuticle or the general body surface, the appendages, or the gills.
- Diagnosis may also be made by bacteriological (isolation, purification and identification) and serological (slide agglutination) methods.

PREVENTION AND CONTROL:

- Maintain good water quality and use nutritionally adequate diets.
- Keep organic load of the water at low levels by removing sediments which harbor high numbers of bacteria.
- Minimize handling and overcrowding and reduce other forms of stress.
- Avoid injuries to the exoskeleton of the shrimps to prevent the development of primary portals of entry.
- Induce molting, which eliminates the condition, but not when underlying tissues are damaged.

6. Internal or Systemic Infections: Luminous Bacterial Disease

CAUSATIVE AGENT: *Vibrio harveyi* and *V. splendidus*

SPECIES AFFECTED: *Penaeus monodon*, *P. merguensis*, and *P. indicus*

STAGES AFFECTED: Eggs, larvae, postlarvae, juveniles and adults

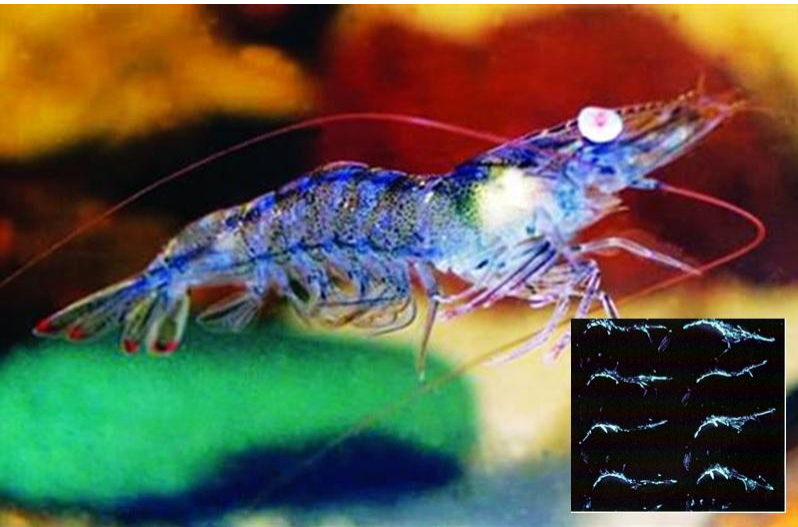


Figure 3-14. Agar culture of the luminous bacteria *Vibrio harveyi*. Photo taken in total darkness

GROSS SIGNS:

- Shrimps become weak and opaque-white.
- Affected shrimps often swim to the pond surface and edges.
- Heavily infected shrimps in tanks and ponds show a continuous greenish glow when observed in total darkness.
- When viewed under the microscope, the hemocoel and internal tissues appear densely packed with active bacteria.

EFFECTS ON HOST:

- The hepatopancreas is the target organ of infection.
- Histopathology shows severe inflammation in and around hepatopancreatic tubules of the entire organ.
- In larger animals, melanized lesions are found in the proximal region of the hepatopancreas.
- These lesions affect the digestive function of the organ as the necrotic parts become non-functional.
- Total necrosis and dysfunction lead to death, while partial dysfunction causes slow growth as not all tubules function in digestion, absorption and storage.
- Systemic infections result in mortality of up to 100%.

DIAGNOSIS:

- The disease may be detected by bacteriological (isolation, purification and identification); histological (demonstration of rod-shaped bacteria in tissues of diseased shrimp); and serological [slide agglutination, fluorescent antibody technique (FAT) and enzyme linked immunosorbent assay (ELISA) using specific antibodies] methods.

- **PREVENTION AND CONTROL:**

- Disinfect incoming water and use filtration equipment to prevent entry of luminous bacteria into the hatchery system.
- Use only previously disinfected water during spawning and rearing.
- Wash eggs.
- Siphon out sediments and debris from the tank bottom.
- Disinfect infected stock first before discarding.
- Wash and disinfect hatchery paraphernalia after each larval rearing period.
- Use microbially mature or aged seawater.
- Apply commercially available probiotics to maintain ecological balance
- within the system.

- Use immunoprophylaxis or vaccination.
- Monitor bacterial population and diversity in the intake and rearing waters of the shrimp pond.
- Apply commercially available probiotics.
- Use low salinity rearing water and reservoirs.
- Practice crop rotation.
- Install greenwater culture system and other system modifications.
- The disease may be prevented by rigorous water management.
- Apply antibiotics and other antibacterial substances only as the last resort.

3. Fungal Diseases

- Many diseases of aquatic animals are caused by organisms that are part of the normal biota of their surrounding environment.
- Fungi and fungal-like organisms (straminipilous organisms) occur in most waters.
- They are either saprobes that colonize decaying organic matter, or parasites, which attack a great variety of aquatic organisms leading to disease outbreaks.
- Among the parasitic species, many are known to be important causative agents of aquatic animal diseases.
- They are generally opportunistic invaders, but once established, are often fatal and difficult to treat.
- Thus, fungi and straminipilous organisms may be problematic pathogens under stressful conditions in an aquaculture system.

1. Saprolegniosis (Saprolegniasis)

CAUSATIVE AGENTS: *Saprolegnia* spp., *Achlya* spp., and *Aphanomyces* spp.

SPECIES AFFECTED: Many freshwater fish (e.g. carps, goldfish)

GROSS SIGNS:

- Formations of white cottony growth on fish eggs and on affected tissues of fish.
- Virtually, any area on the surface of a fish may become infected but it is usually the integuments that are involved.
- Gills, eyes and olfactory pits may also become infected.
- The color of the mycelium may also vary from white to brownish, depending on the color of the particles, which get trapped on the mycelium.

EFFECTS ON HOSTS:

- Once infection is initiated, it is generally progressive and terminal.
- Affected fish become increasingly lethargic, tires more easily and becomes less responsive to external stimuli.
- Loss of equilibrium often occurs shortly before death.
- Histopathologically, evidence of rapid destruction of epidermis (tissue necrosis) with slight inflammatory response can be observed.



DIAGNOSIS:

- Microscopic examination of the cottony growth from the affected tissues will reveal the characteristic hyaline and coenocytic mycelia.
- Usually, numerous sporangia will be present.
- If sporangia are present, one can often make a tentative generic identification.
- However, it must be remembered that other infections (viral, bacterial, protozoan) that are less easy to diagnose might also be present, and that the occurrence of a saprolegnian infection can mask the characteristic signs of these diseases.

PREVENTION AND CONTROL:

- Prevention and treatment of saprolegnian infections of fishes and fish eggs have attracted much attention for a long time now.
- A vast array of chemicals has been tested for effectiveness against these fungi *in vitro*.
- Common chemicals used as chemotherapeutants include bath treatments of either of the following:
 - ✓ Zinc-free malachite green (0.1% on wound and rinse; 67 mg/L for 1 min; 0.2 mg/L for 1 h; 0.1 mg/L indefinite). Cautions: persistent tissue levels; mutagenic; teratogenic; treatment can result in gill damage; more toxic at warm temperatures.

- ✓ Malachite green is considered as the most popular antimycotic agent, being inexpensive and highly effective fungicide. This compound also allows a wide margin of error between therapeutic and toxic dosages.
- ✓ Sodium chloride (22 g/L for 30 min; 30 g/L for 10 min; 1-3 g/L indefinite). This compound is safe to use and inexpensive.
- ✓ Formalin (0.4-0.5 ml/L 30% formaldehyde for 1 h). Inexpensive and popular chemoprophylactic and chemotherapeutic agent.

2. Branchiomycosis (Gill Rot)

CAUSATIVE AGENTS: *Branchiomyces* spp.

SPECIES AFFECTED: Carps, goldfish, eels

GROSS SIGNS:

- Gills become pale with brownish areas due to hemorrhage and thrombosis, or grayish as a result of ischemia.
- Necrotic areas might slough-off at a later stage becoming a focus for saprolegnian infections.



EFFECTS ON HOSTS:

- Fungal hyphae in the gills obstruct the circulation of the blood.
- Necrosis and proliferation of lamellar epithelial cells and lamellar fusions may be observed.
- The disease can appear suddenly and often has a rapid course with losses as high as 30-50% occurring in 2-4 days. Death is due to anoxia.

DIAGNOSIS:

- Microscopic examination will reveal the branched and coenocytic mycelia of the pathogen within the affected gill tissues.

PREVENTION AND CONTROL:

- Various chemicals have been used to treat branchiomycosis, which include:
 - ✓ malachite green (0.3 mg/L for 24 h)
 - ✓ benzalkonium chloride (1-4 ppm active ingredient for 1 h)
 - ✓ copper sulfate (100 ppm for 10-30 min)
 - ✓ sodium chloride (3-5%)
- If an outbreak occurs, feeding of the fish should be stopped and dead fish should be removed from the ponds and buried in a lime pit.
- To help prevent further outbreaks, the pond should be drained, dried-out and disinfected with quicklime.

3. Larval Mycosis

CAUSATIVE AGENTS: *Lagenidium* spp., *Sirolopidium* spp., *Haliphthoros* spp.

SPECIES AFFECTED: All *Penaeus* species, crabs (e.g. *Scylla serrata*)

GROSS SIGNS:

- Sudden onset of mortalities in larval stages of shrimps and crabs.
- Crab eggs are also susceptible for mycotic infection.
- The commonly affected larval stages among shrimp species are the protozoal and mysis stages.

EFFECTS OF HOSTS:

- Progressive systemic mycosis that is accompanied by little or no host inflammatory response can be observed.
- Infection is apparently lethal, accumulating mortality of 20-100% within 48-72 h after onset of infection.

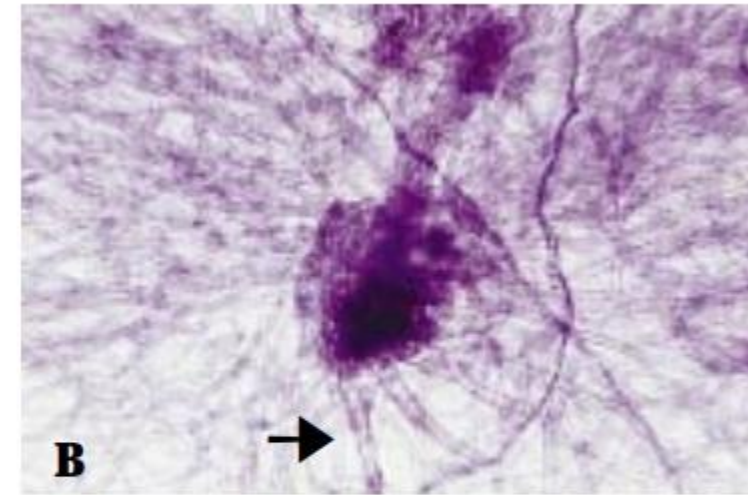
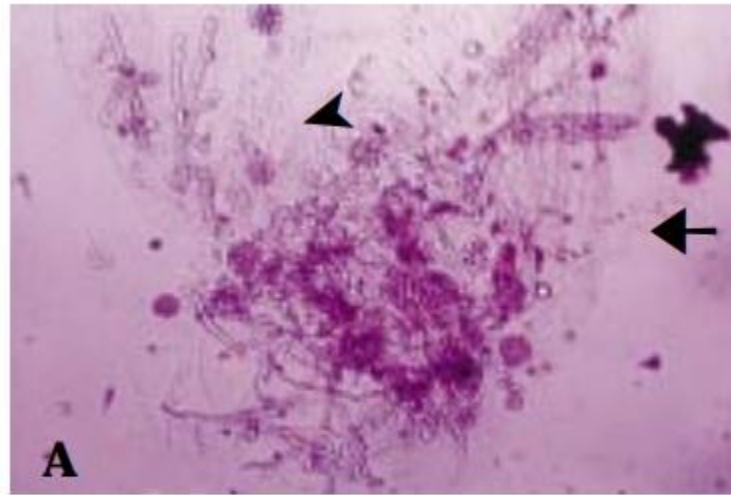


Figure 4-5. *Lagenidium* infection in crustacean larvae. A – Larva of *Penaeus monodon* heavily infested with the fungus (note the mycelia [arrows] completely replacing the body tissues of the larva; and the vesicles [arrowheads] ready to release the zoospores). B – Infected *Scylla serrata* larva in brain-heart-infusion (BHI) broth after 2 days (fresh mount, 40x)

DIAGNOSIS:

- Microscopic examination of affected larvae will reveal extensive, non-septate, highly branched fungal mycelia throughout the body and appendages.
- Specialized hyphae or discharge tubes, with or without terminal vesicles, may be present, and could be the basis for identification of the causative agent.
- Motile zoospores may be observed being released from the discharge tubes in the case of some species.
- Classification of the type of organism causing particular epizootic of larval mycosis is dependent upon the microscopic examination of sporogenesis as follows:
 - ✓ *Lagenidium* – zoospores are released from terminal vesicle.
 - ✓ *Sirlopidium* and *Haliphthoros* – absence of terminal vesicles; zoospores are released through discharge tubes formed by the zoosporangia.

PREVENTION AND CONTROL:

- Disinfection of contaminated larval rearing tanks and chlorination and/or filtration of the incoming water can prevent outbreaks.
- Different antimycotic compounds have been tested *in vitro*.
- Recommended chemicals for therapeutic and prophylactic treatments include the following:
 - ✓ 0.2 ppm Treflan
 - ✓ 1-10 ppm formalin
 - ✓ Egg disinfection with 20 ppm detergent followed by thorough rinsing before hatching

4. Black Gill Disease (Fusarium Disease)

CAUSATIVE AGENT: *Fusarium solani*

SPECIES AFFECTED: All *Penaeus* species

GROSS SIGNS:

- Appearance of “black spots” that preceded mortalities in juvenile shrimps grown in ponds.



EFFECTS ON HOSTS:

- Infection usually starts on damaged tissues such as wounds, gills damaged from chemical treatments or pollutants, and lesions resulting from other disease processes.
- Once infection is established, it is usually progressive with 30% remission rate.
- Lesions may also serve as a route of entry for other opportunistic pathogens.

DIAGNOSIS:

- Microscopic examination of wet mounts of infected tissues will reveal the presence of canoe-shaped macroconidia.
- *Fusarium* spp. are ubiquitous soil fungi.
- Infection may begin at different loci and spread slowly.
- *Fusarium solani* is an opportunistic pathogen of penaeids and are capable of establishing infection in shrimps compromised by other stresses or overcrowding.

PREVENTION AND CONTROL:

- Preventive measures include the elimination of sources of *Fusarium* conidiophores and destruction of infected individuals.
- Several fungicides show promise *in vitro* but none proved to be effective in actual field trials.

5. Aflatoxicosis (Red Disease)

CAUSATIVE AGENT: Aflatoxin produced by *Aspergillus flavus* and other *Aspergillus* spp. which are common contaminants of not-properly stored or expired feeds.

SPECIES AFFECTED: *Penaeus monodon*, other *Penaeus* spp.

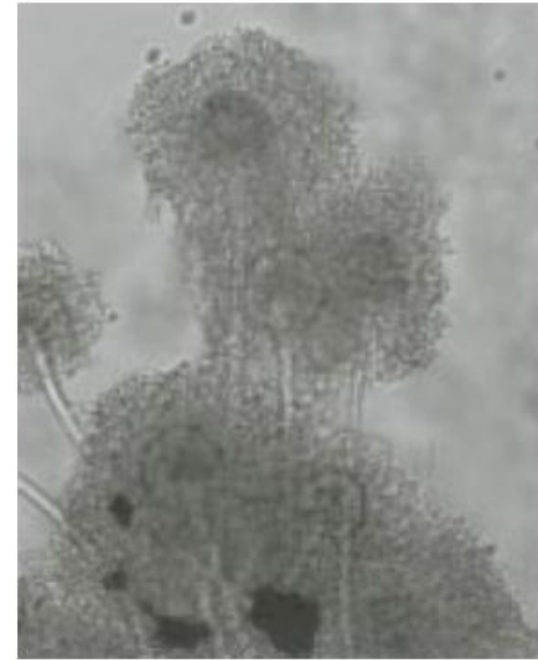


Figure 4-9. Mass of sporangia of *Aspergillus* sp. on contaminated feed particles (fresh mount, 200x)

GROSS SIGNS:

- Yellowish, and eventually reddish discoloration of the shrimp body and appendages can be observed among pond-cultured shrimp juveniles.
- Affected animals become lethargic with weak swimming activity near pond dikes.
- Soft shelling can also be observed.

EFFECTS ON HOSTS:

- Histopathologically, necrosis in the tubule epithelium that proceeds from proximal portion of the tubules to peripheral tubule tips in the hepatopancreas can be observed.
- Growth will be retarded.

DIAGNOSIS:

- Affected shrimps will not survive for more than 30 seconds when collected from the feeding trays.
- There will also be loss of appetite.
- Confirmation is by chemical analysis for the presence of aflatoxin in the suspected feed/ingredient.

PREVENTION AND CONTROL:

- Do not use moldy feeds. Feeds should be properly stored (for not more than 6 months) in dry and well-ventilated areas to prevent, or at least minimize growth of fungal contaminants.

4. Parasitic Diseases

- Aquaculture environments that are suitable for growth and reproduction of cultured animals are also hospitable to potential disease agents such as parasites.
- It is no wonder then that fish mortalities and abnormalities associated with parasites as disease agents are well documented, indicating their importance in aquaculture.
- The study of parasites involves an understanding of certain existing relationships in a particular population.
- **Symbiosis** or “living together” is a relationship that benefits one or both parties. In **commensalism**, no party is harmed and both could live without the other.
- **Mutualism** is a relationship where both parties benefit from each other, and neither could live without the other.

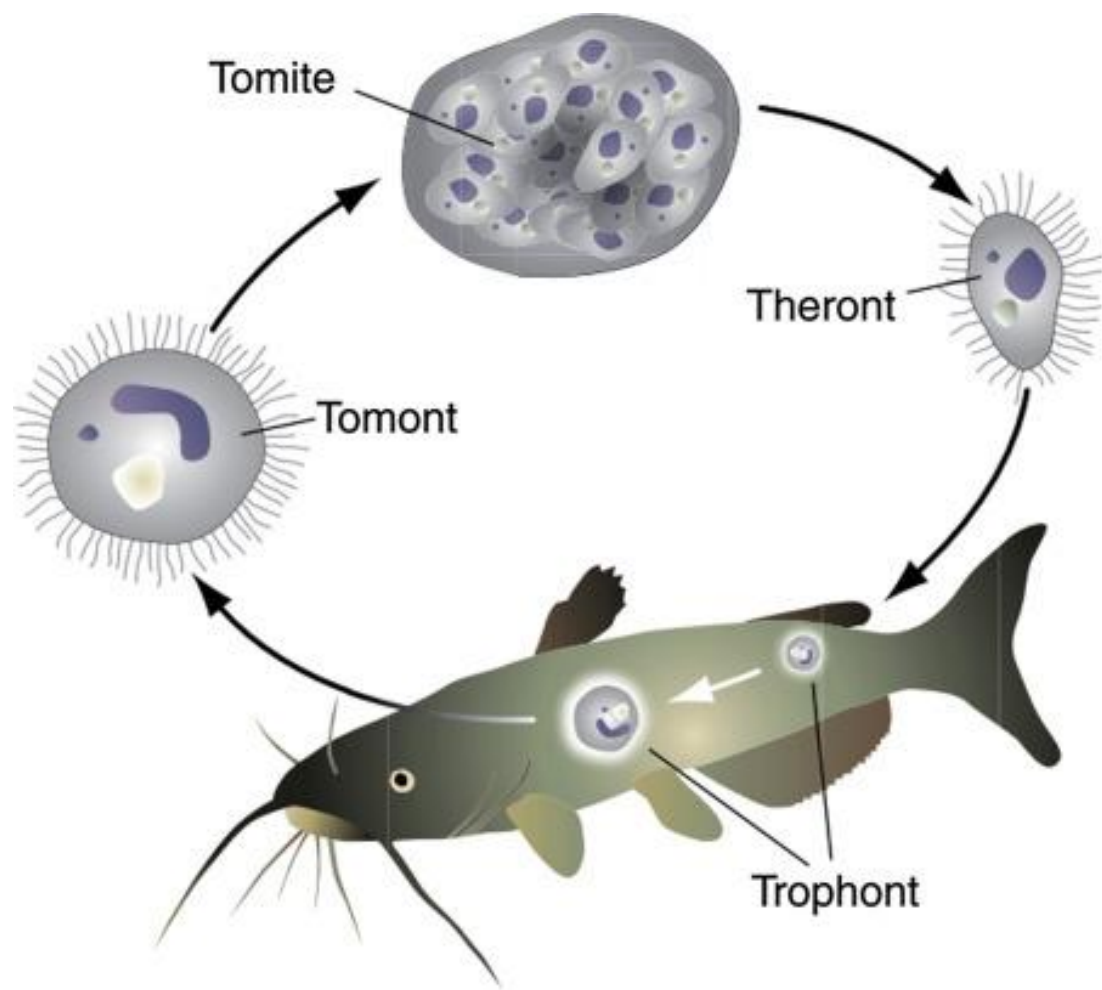
- **Parasitism** is a one-way relationship in which one party (the parasite) depends upon, and benefits from, the other partner (the host), biochemically and physiologically.
- Parasites live in a variety of environments.
- Those that live on the external surfaces (skin, fins, gills) of the host are called ectoparasites, while those found in the internal organs are called endoparasites.

Protozoan Infestations

- Protozoans are unicellular, microscopic organisms with specialized structures for locomotion, food gathering, attachment, and protection.
- They can multiply on or within their hosts.
- ***Ciliates*** have short, fine cytoplasmic outgrowths called cilia as the locomotory organelle.
- They are either attached or motile.
- Ciliates are mainly ectoparasitic.

CAUSATIVE AGENTS:

- *Ichthyophthirius multifiliis* (50-1000 μm diameter) in freshwater.
- *Cryptocaryon irritans* (60-450 μm diameter) in marine and brackishwater.
- The disease is known as Ichthyophthiriasis (“Ich”) or White Spot Disease.



1 Life cycle of Ich. Infective theronts bore through the surface mucus and take up residence within the epithelium of susceptible fish. Theronts differentiate into feeding trophonts that grow and exit the host (as tomonts) within 3 to 8 days depending on temperature. Tomonts divide within the cyst to form hundreds of tomites that differentiate into infective theronts that escape the cyst within 18 to 24 hours at room temperature. Theronts that fail to infect fish die within 1 to 2 days.

SPECIES AFFECTED: Catfish, carp, tilapia, seabass, grouper, snapper

GROSS SIGNS:

- The disease is called “white spot” because of the presence of a few to numerous whitish or grayish spots on the skin and gills of affected fish which are actually nests of these parasites.
- Diseased fish lose their appetite, are lethargic, with dull, opaque or hemorrhagic eyes.
- Heavily infested fish produce a lot of mucus and they rub their body against the substrate or sides of tanks.

EFFECTS ON HOST:

- This disease causes severe epizootic especially in intensive culture systems.
- The parasite may destroy the skin and gills.
- Ulcers may develop in the skin of heavily infested fish and may be the sites of secondary bacterial or fungal infection.
- Occurrence of this parasite is usually associated with a drop in temperature to 28°C.

DIAGNOSIS:

- Encysted (0.10-0.35 mm) organisms appear as white spots on the surface of fish and can be seen by the naked eye.
- Microscopic examination of mucus from the body surface and gill filaments reveals round or oval parasites, propelled by cilia and possessing a horseshoe-shaped macronucleus in the case of *Ichthyophthirius*.

PREVENTION AND CONTROL:

For “Ich”

- Increase water temperature to 30°C for 6 h daily for 3-5 days 0.05% salt solution
- 100 ppm formalin for 1 h for 2-3 days
- 25 ppm formalin + 0.1 ppm malachite green
- Transfer infected stock in dry, parasite-free tanks for 2-3 times at 3 days interval

For *Cryptocaryon*

- 0.5 ppm CuSO₄ and 25 ppm formalin for 5-7 days, then transfer to dry, parasite-free tanks for 2 times at 3 days interval.

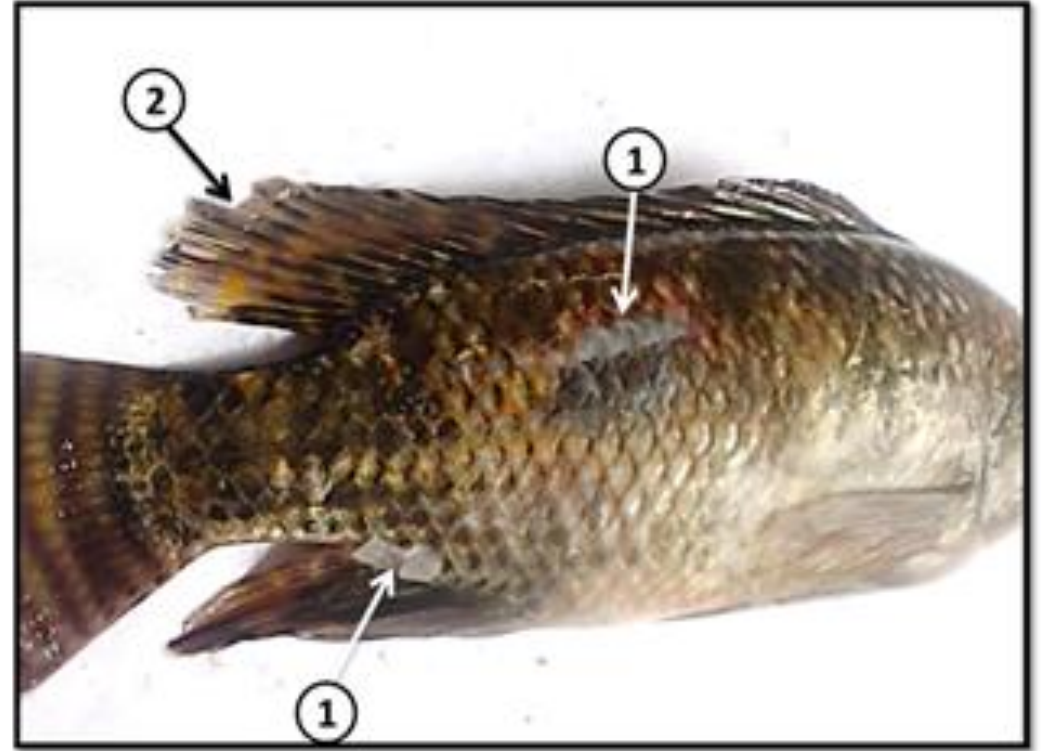
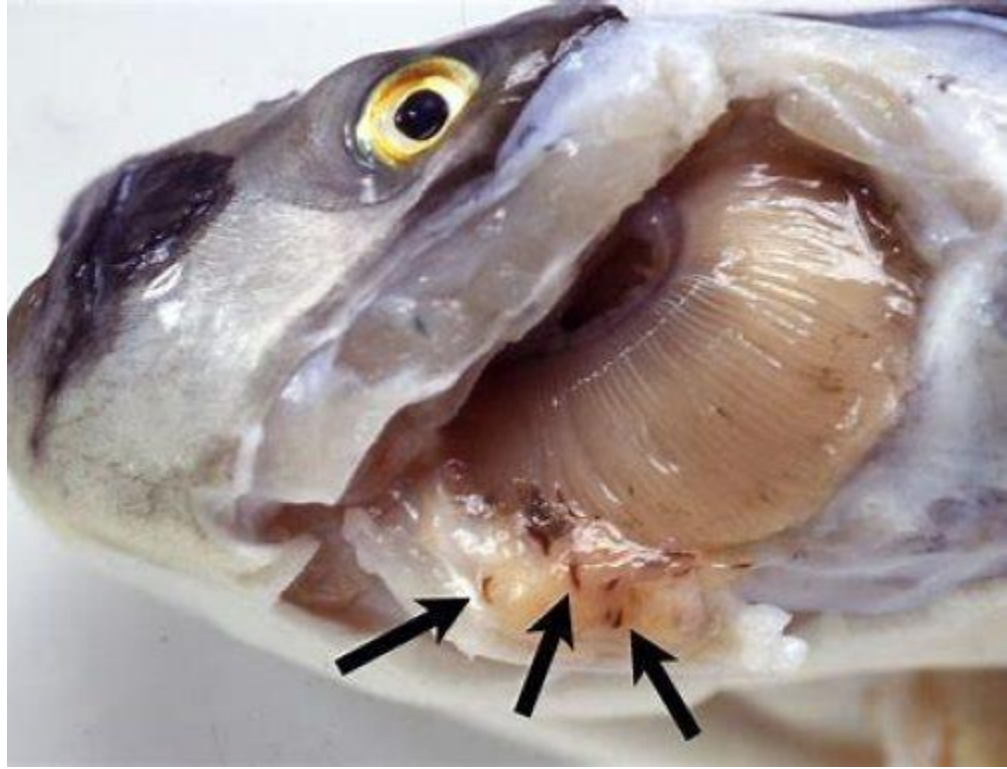
Protozoan Infestations continued...

CAUSATIVE AGENTS: *Trichodina* (45-78 µm diameter), *Trichodinella* (24-37 µm diameter), *Tripartiella* (up to 40 µm diameter)

SPECIES AFFECTED: Carp, tilapia, milkfish, seabass, mullet, siganid, grouper, snapper

GROSS SIGNS:

- The parasites are attached mainly on the gills and skin of the host.
- Affected fish appear weak with excessive mucus production and with frayed fins.



EFFECTS ON HOST:

- Excessive numbers of the parasite on the skin and gills of infested fish may interfere with respiration.
- High mortality was observed among young fish.
- The adhesive disc can cause direct damage to the branchial epithelium resulting in gill lesions.

DIAGNOSIS:

- Microscopic examination of wet mounts of gill filaments and scrapings from skin show saucer-shaped organisms, surrounded by cilia around its perimeter.

PREVENTION AND CONTROL:

- 2-3% salt solution for 2-5 min for 3-4 days (carp fry)
- 100% freshwater bath for 1 h for 3 days
- 100 ppm formalin+10 ppm Acriflavin for 1 h for 3 days

Monogenean Infestations

- Monogeneans are ectoparasitic flatworms, <1-5 mm long, with posterior organ of attachment called haptor armed with hooks and/or suckers.

CAUSATIVE AGENTS: *Gyrodactylus*, *Dactylogyrus*

SPECIES AFFECTED: Catfish, carp, tilapia, seabass, grouper, snapper

GROSS SIGNS:

- Parasite attaches on gills, fins and body surface of fish.
- Affected fish have pale skin and gills with increased mucus production, frayed fins, and the cornea may become opaque.



EFFECTS ON HOST:

- Heavy infestation may result to hyperplasia of the epithelial cells in the skin.
- Extensive damage to the gill epithelium may affect normal respiration.
- Heavy infestations may result in mortality.
- Conditions of low oxygen levels may increase mortality rates.
- Often associated with vibriosis.

DIAGNOSIS:

- Gross and microscopic examination of gills and body surface of freshly sacrificed fish.

PREVENTION AND CONTROL:

- Maintain optimum stocking density and adequate feeding
- 5% salt solution for 5 minutes
- Freshwater bath for 1 h for 3 days
- 100 ppm formalin for 1 h for 3 days
- 150 ppm hydrogen peroxide for 30 minutes

Crustacean Infestations: *Argulus* (fish louse) Infestation

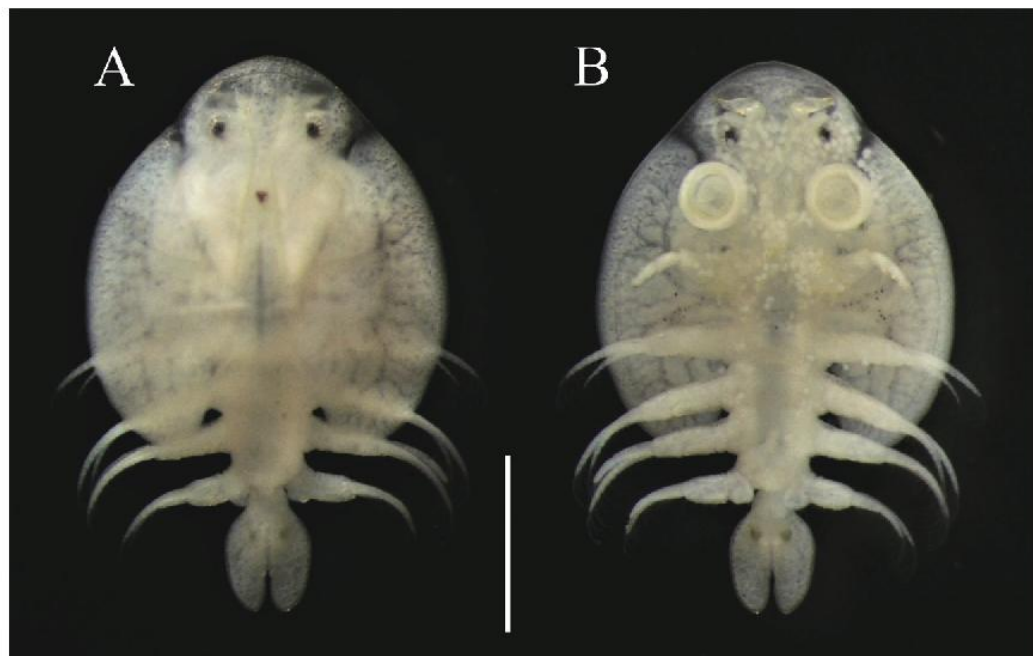
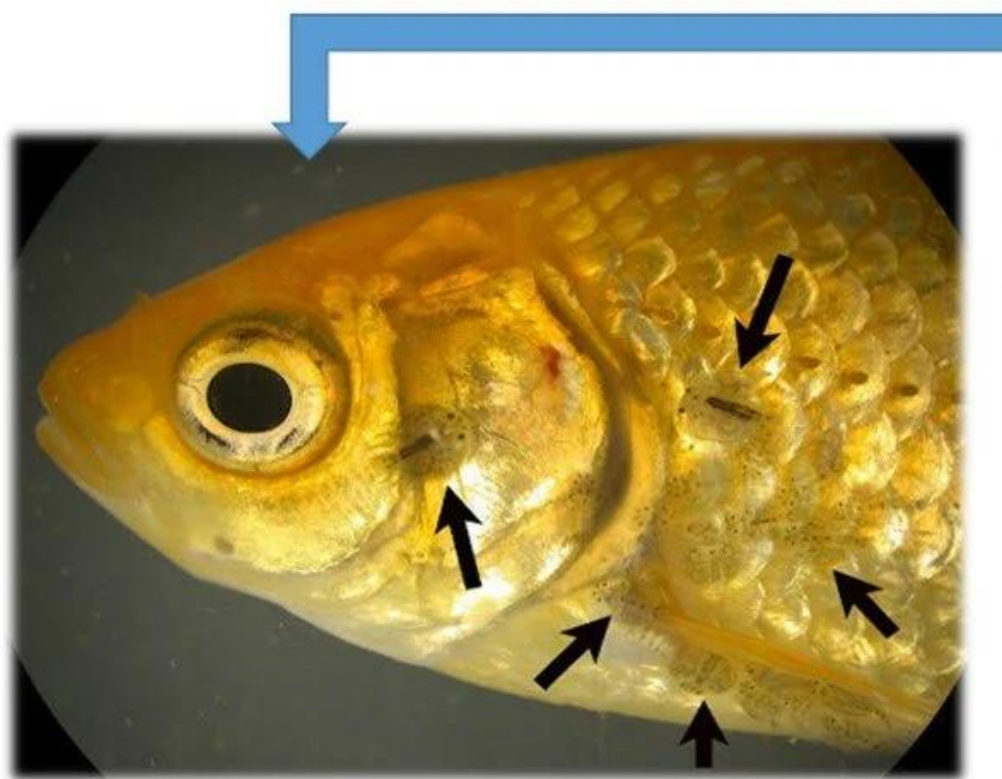
- Crustacean parasites have segmented bodies covered by shell with jointed appendages.

CAUSATIVE AGENT: *Argulus*; Female, 6-6.5 mm; male, 2-3 mm.

SPECIES AFFECTED: Tilapia, milkfish, mullet, carp, snakehead, catfish

GROSS SIGNS:

- Parasite attached to the skin, fins, buccal or opercular mucosa with two conspicuous black spots.
- Attachment area hemorrhagic or ulcerated.



1. *Argulus ignonicus*, immature female, NSMT-Cr 25568, from the caudal fin of *Opsariichthys platymus* in a f

EFFECT ON HOST:

- Heavy infestation may result in mortalities.
- Wounds may become necrotic and ulcerated, paving the way for secondary bacterial infection.

DIAGNOSIS:

- Gross and microscopic examination of host fish.

PREVENTION AND CONTROL:

- Use of filters
- 5 ppm KmnO_4 , 3-5 min
- 1 ppm Dipterex, 3-6 hours

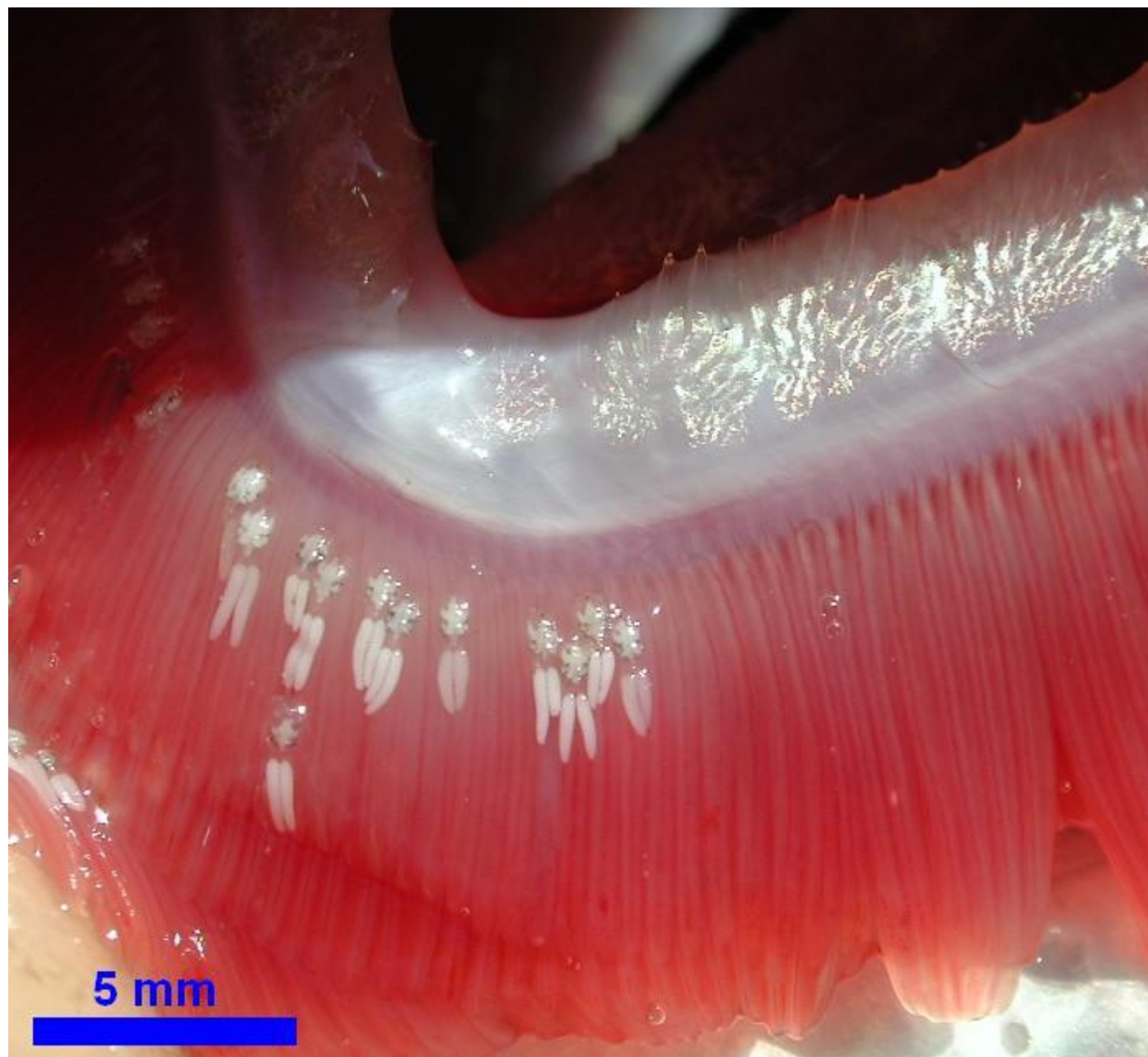
Crustacean Infestations: Ergasilid Infestation

CAUSATIVE AGENT: *Ergasilus*; 0.9-1.5 mm long

SPECIES AFFECTED: Tilapia, carp, goby, seabass, grouper, mullet

GROSS SIGNS:

- Parasite attaches to the gills and body surface of the host.
- White to dark brown copepods, <2 mm long, some with 2 white elongated egg sacs are firmly attached to the gills.
- Affected fish have emaciated body.



EFFECTS ON HOST:

- Destruction of the gill filaments resulting to respiratory dysfunction.
- Infestation of 10-70 parasites per fish can lead to 50% fish loss. Secondary bacterial infection is common.

DIAGNOSIS:

- Gross and microscopic examination of host fish.

PREVENTION AND CONTROL:

- 0.15 ppm Bromex bath for 1 week
- 0.25-0.5 ppm Dipterex bath for 24 hours

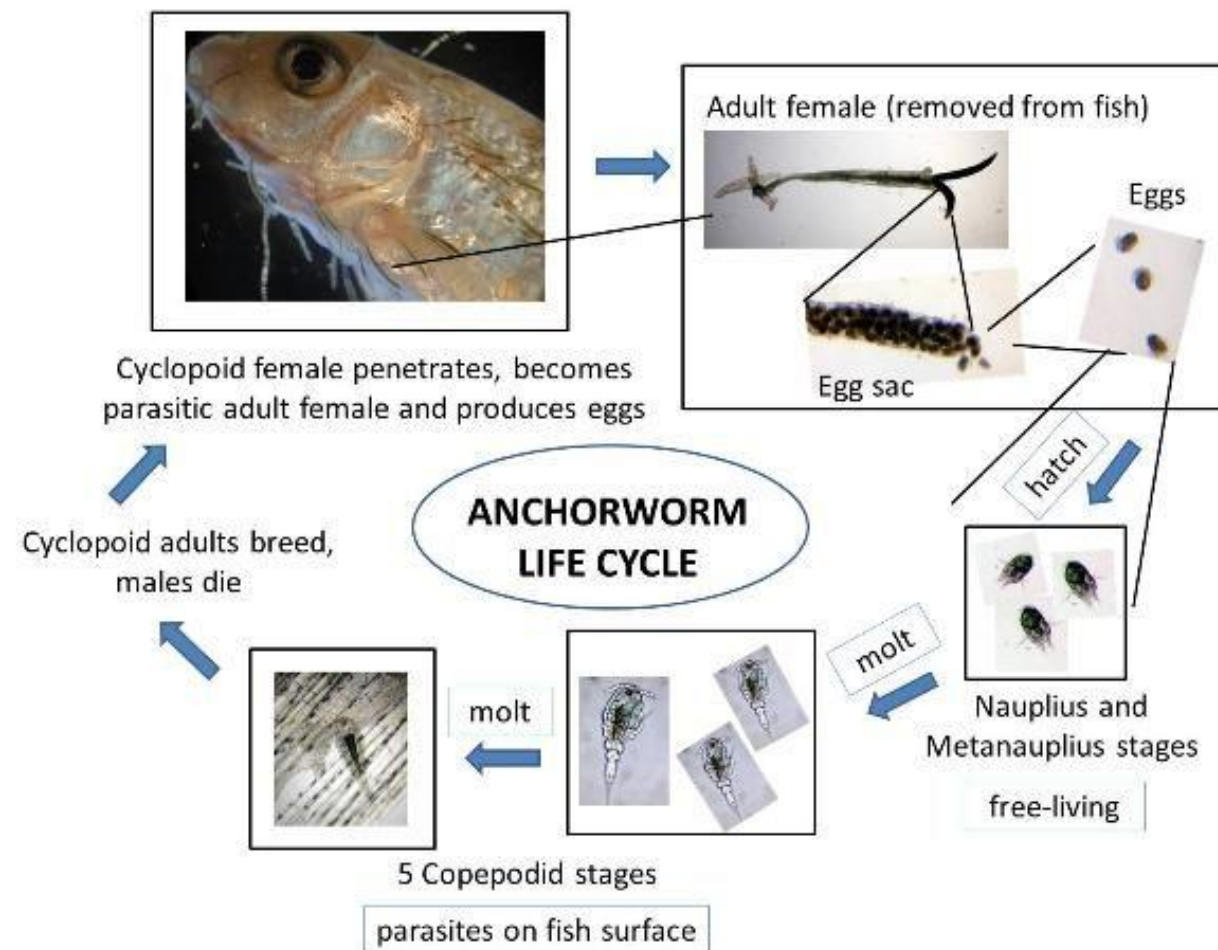
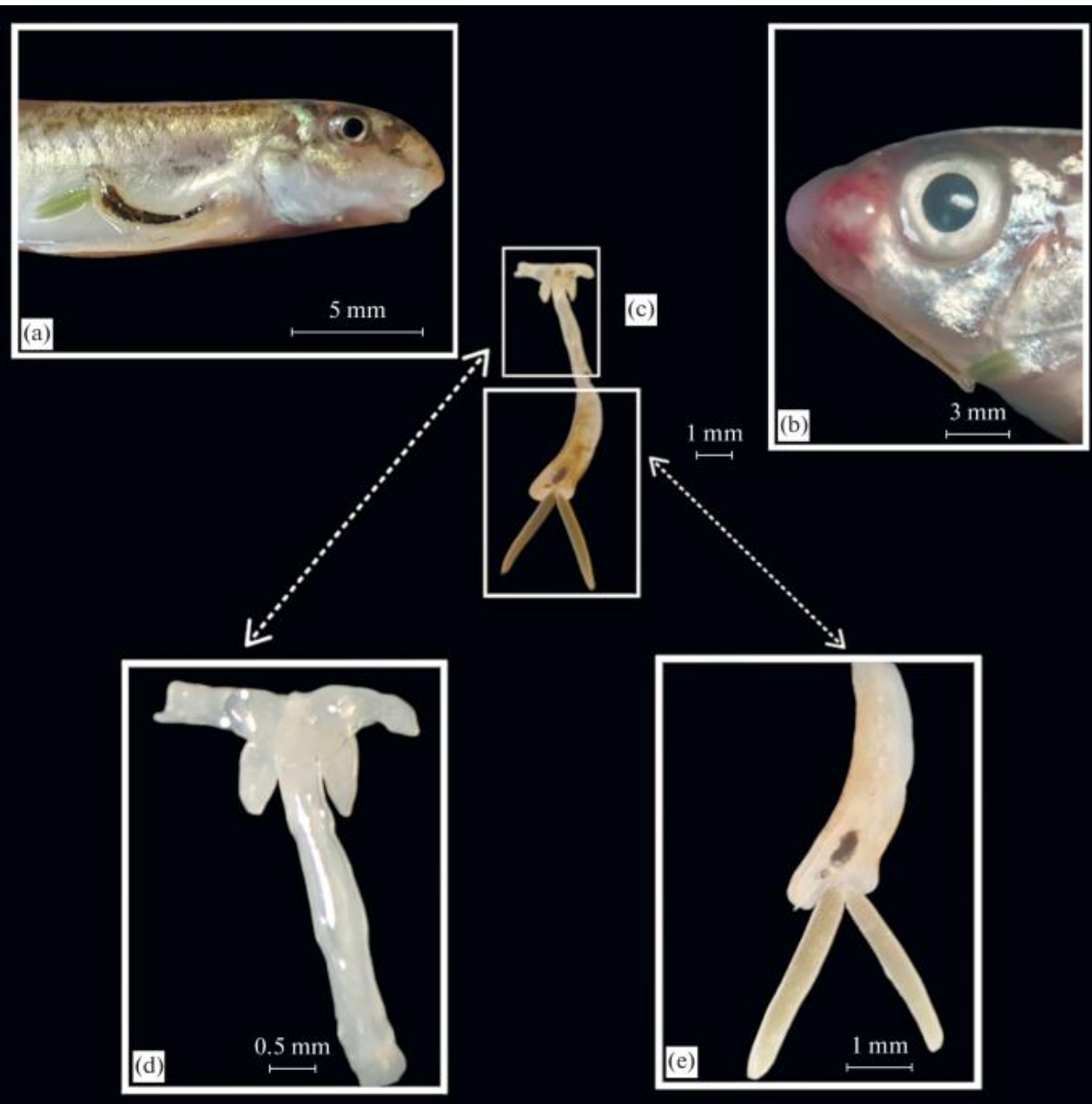
Crustacean Infestations: Lernaeid Infestation

CAUSATIVE AGENT: *Lernaea* or “anchor worm”; 1.2-1.4 mm long

SPECIES AFFECTED: Milkfish, carps, tilapia, snakehead, goby

GROSS SIGNS:

- Parasites observed protruding from nostrils, skin, bases of fins, gills, buccal cavity.
- Heavy infestation can result to loss of scale and skin ulcerations.



EFFECTS ON HOST:

- The skin and muscle are swollen.
- Ulcers may develop and result to muscle necrosis.
- The site of attachment may be a portal of entry for secondary microbial infection.
- Parasite attachment on the head may cause twisting and deformation of jaws.
- Affected fish suffer from serious weight loss.
- Mass mortality is not uncommon.

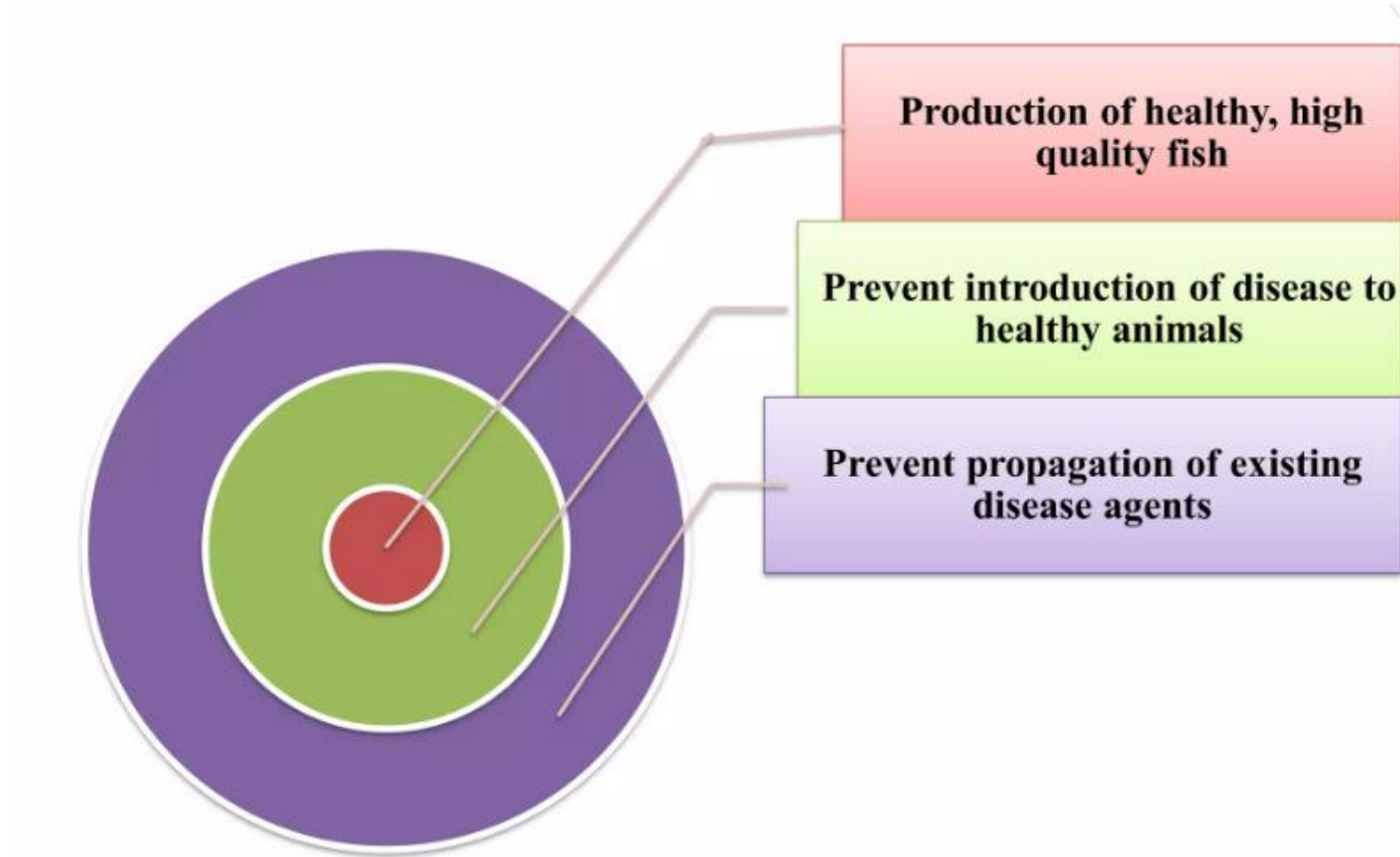
DIAGNOSIS:

- Gross and microscopic examination of host fish.
- The body shape and attachment organ of the parasite have earned it its name – “anchor worm.”

PREVENTION AND CONTROL:

- 3-5% salt solution for control of larval stages
- Adult stages may be eliminated through pond drying and liming

Fish Health Management Goals



Principles of Fish Health Management

- Fish health management in aquaculture is based on prevention, early detection, and proper control of diseases to ensure sustainable production.
- The main principles include;
 1. Prevention is Better Than Cure
 2. Good Biosecurity Practices
 3. Proper Water Quality Management
 4. Early Diagnosis and Regular Monitoring
 5. Stress Reduction
 6. Appropriate Treatment and Responsible Drug Use
 7. Record Keeping
 8. Vaccination

1. Prevention is Better Than Cure

- Maintain optimum water quality (temperature, dissolved oxygen, pH, ammonia).
- Avoid overcrowding and reduce stress.
- Use high-quality, nutritionally balanced feed.
- Purchase healthy, disease-free seed from reliable hatcheries.

2. Good Biosecurity Practices

- Restrict movement of people, equipment, and vehicles.
- Disinfect equipment regularly.
- Use footbaths and hand sanitation.
- Prevent entry of wild fish and other carriers.
- Isolate (quarantine) new stock before introducing them to ponds or tanks.

3. Proper Water Quality Management

- Regular monitoring of physical, chemical, and biological parameters.
- Maintain stable environmental conditions.
- Ensure proper filtration and water exchange.

4. Early Diagnosis and Regular Monitoring

- Observe fish behaviour daily (feeding response, swimming pattern).
- Check for abnormal signs (lesions, discoloration, gasping).
- Conduct laboratory examination when necessary.

5. Stress Reduction

- Handle fish gently during grading, transfer, and harvesting.
- Avoid sudden changes in temperature or salinity.
- Maintain appropriate stocking density.

6. Appropriate Treatment and Responsible Drug Use

- Use chemicals and antibiotics only when necessary.
- Follow recommended dosages and withdrawal periods.
- Avoid misuse to prevent antimicrobial resistance.

7. Record Keeping

- Maintain records of water quality, feeding, growth, treatments, and mortality.
- Helps in early detection of recurring problems and better decision-making.

8. Vaccination (Where Applicable)

- Use vaccines for certain bacterial and viral diseases in intensive systems.

- The key principle of fish health management is preventive management through good husbandry, biosecurity, and water quality control, rather than relying only on treatment after disease outbreaks occur.

A top-down view of a desk setup. In the center is a spiral-bound notebook with a white cover. The words "THANK YOU!" are printed on the notebook in a large, bold, sans-serif font. "THANK" is in black, and "YOU!" is in black with a red exclamation mark. To the top left of the notebook are a pair of gold-rimmed glasses. To the right is a silver and black ballpoint pen. To the bottom left is a small white pot containing a green succulent plant. The entire scene is set on a light-colored, textured wooden surface.

**THANK
YOU!**